Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

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I. WORK DONE THIS WEEK:-

In Week 3 we built a training-free GLCM+LBP pipeline on the TILDA database, but the evaluation was only qualitative. Our own limitations list said that each reference came from a single normal image that was also evaluated against it, that no ROC or precision/recall against ground-truth masks was reported, and that the validation used only one defect image per class. This week we therefore concentrated on turning the project into a properly *measurable* one: we moved to a dataset with pixel-level ground-truth masks, fixed a train/test protocol, compared several scoring methods using training data only, froze one configuration, and evaluated it once on held-out images.

A. Dataset and evaluation protocol

We switched to the AITEX fabric image database (AFID) [6], because every defective image is supplied with a ground-truth mask, which allows pixel-level precision, recall, F1 and IoU. The data consist of defective images, their masks, and defect-free images. Image–mask pairs are listed in a pairing manifest CSV with 106 rows; 105 are "matched" with matching image/mask dimensions and one has no mask, so only the 105 valid pairs are used (Steps 5–6). There are 107 mask files because some images have more than one mask, and 141 defect-free images, all of size 256×4096 pixels (Step 9). A sample image–mask pair was loaded and validated (Step 6) and the mask was overlaid on the image to confirm alignment (Step 7, Fig. 1); in this sample the defect covers only 0.133% of the pixels. All 141 defect-free images loaded correctly, with mean gray level 125.7 on average (Step 9), and a normal/defective comparison was made (Step 10, Fig. 1); the defect in that example is only 31 pixels.

To avoid the optimistic estimates of Week 3, we created a reproducible image-level split (80%/20%, seed 42, Step 8). A split stratified by defect code was attempted but is impossible because one defect code has only a single image, so a reproducible unstratified split was used. The 105 valid pairs became 84 training and 21 test images with no image overlap (checked by an assertion in the code). The training set has 12 defect codes, while the test set contains 7 of them (codes 2, 6, 19, 22, 23, 25, 30; 1 to 8 images each). The test images were not used for any design decision; the method, patch size and threshold were all chosen on the training images and then frozen (Step 34) before the single test run (Step 35).

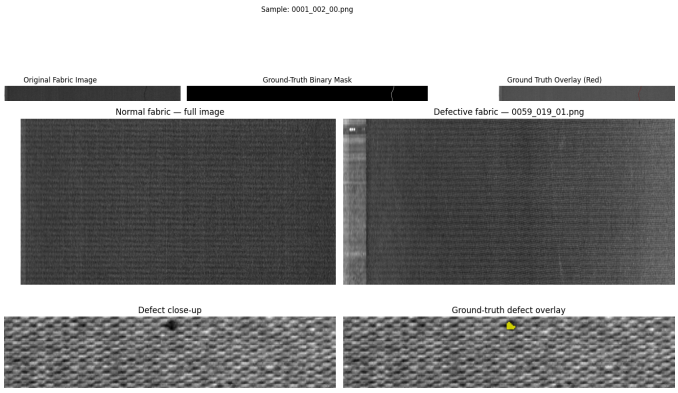


Fig. 1. AITEX samples: original image, ground-truth mask and overlay (Step 7, top), and normal vs. defective fabric with close-up of a small dark defect (Step 10, bottom).

TABLE I
DATASET AND SPLIT SUMMARY (STEPS 5, 8, 9)

Item	Count
Manifest rows (defective images)	106
Valid image–mask pairs	105
Defect-free images (256×4096)	141
Training images	84
Held-out test images	21
Train/test overlap	0

B. Patch features

Each image is divided into non-overlapping square patches and a 26-dimensional feature vector is computed per patch (Step 11). The GLCM part uses 16 gray levels, distance 1 and angles $0°, 45°, 90°, 135°$, with contrast, dissimilarity, homogeneity and energy per angle (16 values). The LBP part is a normalized 10-bin uniform LBP histogram ($P = 8$, $R = 1$). Compared with Week 3 we reduced the quantization from

32 to 16 levels and use dissimilarity in place of correlation. Features were extracted from the defect-free images using 32×32 patches, giving 144,384 normal reference patches (Step 12, matrix of shape 144384×26), and from the training images with a defect label for each patch: a patch is positive if it contains at least one ground-truth defect pixel (Steps 14E–15).

C. Anomaly-scoring methods compared on training data

(a) Nearest-normal distance (Steps 13, 16–18). Features are standardized with a scaler fitted only on normal patches, and the score of a patch is its Euclidean distance to the nearest normal patch (KD-tree). This replaces the single-image reference of Week 3, which failed when one textile's reference was applied to another. On the training patches (1,222 positive and 84,714 negative) this gives a pooled ROC-AUC of only 0.6584 and an average precision of 0.0196 (Step 17, Fig. 2); the median score is 0.7501 for defect patches and 0.4503 for normal ones.

(b) Local texture deviation (Steps 19–21). Each patch is compared with its spatial neighbours (a 5×5 neighbourhood of patches, excluding itself):

$$s_i = \underset{f}{\text{median}} \; \frac{|x_{i,f} - m_f|}{\max(1.4826\,\text{MAD}_f,\ \epsilon_f)}, \tag{1}$$

where m_f and MAD_f are the neighbourhood median and median absolute deviation of feature f. This needs no external reference and adapts to each image's own background. On 83 images it raised the mean per-image AUC from 0.5796 (nearest-normal) to 0.7019 and was better on 52 of them (Step 20, Fig. 3a).

(c) Patch size (Steps 22–23, 32). Local-deviation AUC was compared for 16×16 and 32×32 patches (Step 22): the mean per-image AUC rose from 0.7019 to 0.7548 (median 0.7195 to 0.7684) on 83 images. 16×16 patches were therefore used from Step 23 onward, and an 8×8 vs. 16×16 comparison for the intensity score below was made in Step 32.

(d) Threshold sweep and a failure case (Steps 24–27). Percentile thresholds (90th–99.5th) were swept on pooled training scores (Step 24) and a pixel mask was rebuilt from patch predictions for one image (Step 25). Many AITEX defects are tiny (the example in Step 10 has 31 pixels), so a single defect barely changes the texture statistics of its patch. We inspected the score map around such a defect (Step 26) and tested an intensity cue, comparing the defect-containing patch with its surroundings (Step 27); for this image the dark-contrast AUC (0.994) was higher than the texture AUC (0.893) (Step 29).

(e) Local dark contrast (Steps 27–28). Let $\mu_{i,j}$ be the mean gray level of patch (i, j). The score is the difference to the local median of the patch-mean grid,

$$D_{i,j} = \text{med}_{5\times5}(\mu)_{i,j} - \mu_{i,j}, \tag{2}$$

so a positive value means the patch is darker than its neighbourhood. This is an adaptive baseline: the reference level is estimated locally in every image and is not a global constant.

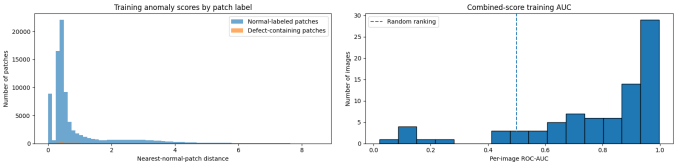


Fig. 2. (a) Nearest-normal anomaly scores by patch label (Step 17). (b) Per-image ROC-AUC of the combined score (Step 29).

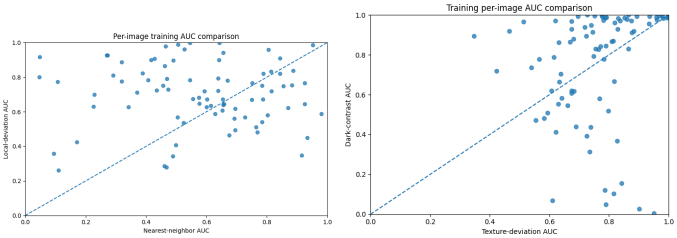


Fig. 3. Per-image training AUC: (a) local deviation vs. nearest-normal (Step 20); (b) dark contrast vs. texture deviation (Step 28). Points above the dashed line favour the y-axis method.

Its mean per-image AUC (0.7526, median 0.8678) is similar to the texture score (0.7548), but it is better on 53 of 83 images (Step 28, Fig. 3b).

(f) Combined score (Steps 29–31). Texture and dark-contrast scores are robustly normalized per image, $z = (s - \text{med})/(1.4826\,\text{MAD})$, and averaged with equal weights, $S = 0.5\,z_{\text{tex}} + 0.5\,z_{\text{dark}}$. The combined score reached the highest mean AUC, 0.7788 (median 0.8828), and beat the texture score on 62 of 83 images (Step 29, Fig. 2b). A threshold sweep (Step 30) and a pixel-level training evaluation (Step 31) were run. The three 16×16 methods were then compared with the same 98th-percentile rule (Step 33).

D. Frozen configuration and held-out test

With the same 98th-percentile rule, dark contrast gave the best patch-level F1 on the training data (0.0833, against 0.0491 for texture deviation and 0.0730 for the combined score; Table III), even though the combined score has the highest mean AUC. We therefore chose, as the final candidate, the local dark-contrast detector with 16×16 patches and the threshold set at the 98th percentile of pooled training scores, $T = 5.3203$ gray levels (Step 34). On the 21 test images (Step 35) every patch with $D \geq T$ is marked, patch decisions are expanded to 16×16 pixel blocks, and precision, recall, F1 and IoU are computed against the ground-truth masks per image and pooled over all images. No threshold or parameter was changed on test data. Predicted masks and a per-image metrics CSV were saved. Representative worst, median and best cases (Step 36, Fig. 4) and a breakdown by defect code (Step 37) were produced, and Step 38 prints the final summary (Tables IV and V).

E. Observations

The scoring variants rank patches reasonably well (mean per-image AUC 0.75–0.78), but turning scores into a mask

TABLE II
PER-IMAGE TRAINING ROC-AUC OF THE SCORING VARIANTS (83 IMAGES)

Score (patch size)	Step	Mean	Median
Nearest-normal (32)	20	0.5796	0.6019
Local texture dev. (32)	20/22	0.7019	0.7195
Local texture dev. (16)	22	0.7548	0.7684
Dark contrast (16)	28	0.7526	0.8678
Combined (16)	29	0.7788	0.8828

TABLE III
PATCH-LEVEL COMPARISON ON TRAINING DATA, SAME 98TH-PERCENTILE RULE (STEP 33)

Method	Thr.	Prec.	Rec.	F1
Dark contrast	5.3203	0.0614	0.1295	0.0833
Texture deviation	1.4614	0.0362	0.0762	0.0491
Combined	2.2731	0.0538	0.1133	0.0730

TABLE IV
HELD-OUT TEST RESULTS, FROZEN DETECTOR, 21 IMAGES (STEPS 35, 38)

	Prec.	Rec.	F1	IoU
Mean per image	0.0203	0.3726	0.0326	0.0189
Median per image	0.0	0.0	0.0	0.0
Pooled pixels	0.0164	0.3480	0.0313	0.0159

Images with IoU $= 0$: 11/21 (52.4%) IoU > 0: 10/21 (47.6%)

TABLE V
HELD-OUT TEST RESULTS BY DEFECT CODE (STEP 37)

Code (images)	Prec.	Rec.	F1	IoU
2 (2)	0.1752	0.7770	0.2716	0.1627
6 (3)	0.0163	0.1597	0.0296	0.0155
22 (2)	0.0034	0.0929	0.0065	0.0033
30 (3)	0.0023	0.6667	0.0046	0.0023
19 (8)	0.0016	0.4506	0.0031	0.0016
23 (2)	0.0	0.0	0.0	0.0
25 (1)	0.0	0.0	0.0	0.0

is much harder. The 98th-percentile rule flags about 2% of patches, whereas defects cover a tiny fraction of the pixels, so on the held-out set pooled precision is only 0.0164 with recall 0.348, and the median per-image IoU is 0 (11 of 21 images have no overlap). The best defect code is 2 (mean IoU 0.1627), and the best single image is a thin dark vertical line (IoU 0.2429, Fig. 4, bottom). These numbers are weak, and we report them as they are: the current detector is a baseline, not a finished solution.

The failures have two clear causes. First, dark contrast can only find defects that are darker than their surroundings. The images with the lowest dark-contrast AUC in Step 28 are mostly defect code 6 (AUC 0.0035 to 0.12), and in the held-out example 0016_006_02 the defect is a bright vertical line, so it is missed while many false patches are flagged (Fig. 4, middle). Second, very small defects (Fig. 4, top) cover only a few pixels of a 16×16 block, and block-shaped predictions limit the attainable IoU even when the right block is found. The test set is also small and only partly covers the defect codes (Table V), so the per-code numbers are indicative only.

The final detector is based on a local intensity statistic; the GLCM and LBP descriptors were used in the nearest-normal, local-deviation and combined variants. The adaptivity lies in the locally estimated background (the median of neighbouring patch means) rather than in the threshold, which is one global constant chosen on training data. A per-fabric or per-image threshold remains future work.

Our Week 3 pipeline is summarized as: Fabric Image $\rightarrow$ Windows $\rightarrow$ GLCM/LBP $\rightarrow$ z-score vs. one normal image $\rightarrow \mu + k\sigma$. The current pipeline is: Fabric Image $\rightarrow 16 \times 16$ patches $\rightarrow$ local dark-contrast score $\rightarrow$ frozen training threshold $\rightarrow$ pixel mask $\rightarrow$ metrics against ground truth.

II. WORK TO BE DONE NEXT WEEK:-

Next week we will improve localization quality and prepare the final deliverables. Patch-level decisions give blocky masks, so we plan to try overlapping windows and a pixel-level refinement inside flagged patches (for example local thresholding, morphological clean-up and connected-component filtering), and to measure whether IoU improves over the block masks.

The present detector only finds patches that are darker than their surroundings, and the results show that bright defects (for example code 6) are missed. We will extend the score to two-sided contrast (absolute deviation from the local median) and combine it with the texture features in a better way than the equal-weight average, since texture deviation was the stronger cue on exactly those images.

We will also test per-fabric (or per-image) thresholds to make the decision rule genuinely adaptive, and report results by defect code with a clear note that the test set is small (21 images). Finally, we will organize the code and results in the GitHub repository, clean up the notebook (removing the recovery cells), and write the final report and presentation.

III. LITERATURE REVIEW:-

This week the literature review was used to interpret our results rather than to find new methods. Papers [1], [2] and [3] use LBP, GLCM and their combination to describe fabric texture; our nearest-normal and local-deviation scores build directly on this kind of feature vector. Our observation that texture statistics alone do not always highlight very small defects suggests that other cues, such as local intensity contrast, should be fused with them, which is also the motivation for decision/score fusion in [4]. The comparison of GLCM with Gabor filters in [5] points to filter-bank features as a possible extension if texture-only scoring remains weak.

For the evaluation protocol, we adopted the AITEX database [6], which provides pixel-level ground-truth masks, so that precision, recall, F1 and IoU can be computed instead of only counting flagged windows. We also follow the standard

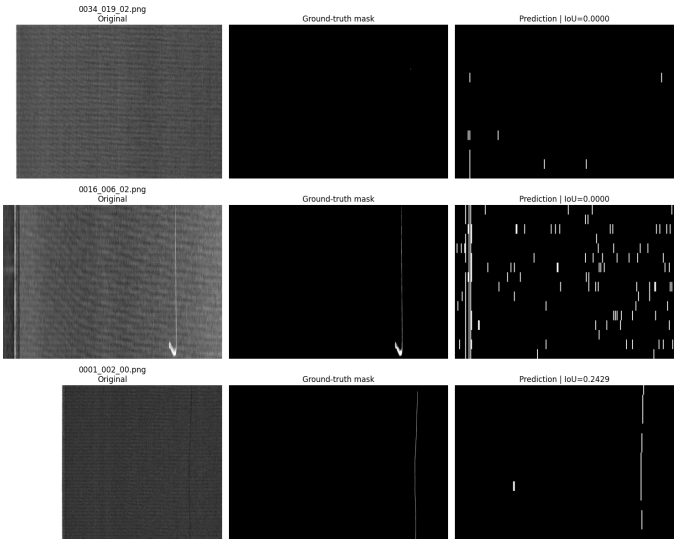


Fig. 4. Held-out test cases (Step 36): two failures (IoU 0) and the best case (IoU 0.2429). Columns: image, ground truth, prediction; images are stretched horizontally to fit.

practice of choosing parameters on training data and reporting results on a held-out set.

IV. CHALLENGES FACED:-

The first challenge was the change of dataset. Image–mask pairing is not trivial: there are 107 mask files for 106 defective images and one image has no mask, so a manifest, a loader that merges masks and a size check were needed. The defect-code distribution is also very uneven (one code has a single image), which made a stratified split impossible and left the test set with only 7 of the 12 codes.

The second was the very different nature of the defects. Some are tiny (31 pixels in one example), so a 32×32 patch averages them out; others are bright rather than dark lines. This is why we studied smaller patches and added an intensity-contrast score, which in turn fails on bright defects.

The third was keeping the evaluation honest: the patch size, method and threshold were selected only on training images, and the test set was used once. Patch-level labels (positive if any defect pixel lies inside) and block-shaped predictions also limit the pixel-level IoU that can be reached for thin defects, and the held-out result (mean IoU 0.0189) is low.

Finally, long feature-extraction runs were interrupted by Colab runtime resets, so we added restore and checkpoint cells (Steps 14C–14G) to resume without recomputing everything.

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